

Dumpster Diver — BOV Pilot Spec Sheet (F1–F4)

Target pH 5.8–6.0 • Viscosity window 250–600 cP • Filter 100–200 µm • Preservative: Optiphen Plus 1.0% or Phenoxyethanol 0.8%

F1 — Baseline Foaming Cleanser

Ingredient	% (w/w)	1 L Batch (g)
Deionized Water	q.s. to 100%	
SCI (Sodium Cocoyl Isethionate)	4.0%	40 g
CAPB (Cocamidopropyl Betaine)	6.0%	60 g
Capryl Glucoside	2.0%	20 g
Polysorbate 20	1.0%	10 g
Vegetable Glycerin	2.0%	20 g
Sodium Phytate	0.20%	2 g
Peppermint Essential Oil	0.30%	
Preservative (Optiphen Plus 1.0% or Phenoxyethanol 0.8%)	see note	10 g (Optiphen) / 8 g (Phenoxy)
Citric Acid (10% solution) to pH	5.8–6.0	

Goal: Simple, clear, stable; good foam; VOC-free; BOV-ready. **pH:** 5.8–6.0 **Viscosity:** ≈250–400 cP

Notes: Primary foam from SCI + CAPB; add preservative <40°C; filter 100–200 µm before BOV; de-gas.

F2 — Boosted Cleaning (Hydrotrope + Co-solvent)

Ingredient	% (w/w)	1 L Batch (g)
Deionized Water	q.s. to 100%	
SCI	3.5%	35 g
CAPB	6.0%	60 g
Capryl Glucoside	2.0%	20 g
SXS (Sodium Xylene Sulfonate, 40%)	2.0%	20 g
DPG (Dipropylene Glycol)	2.0%	20 g
Sodium Gluconate (or Phytate 0.2%)	0.30%	3 g
Polysorbate 20	1.0%	10 g
Vegetable Glycerin	2.0%	20 g
Peppermint EO	0.30%	3 g
Preservative	as F1	10 g (Optiphen) / 8 g (Phenoxy)
Citric Acid (10% solution) to pH	5.8–6.0	

Goal: Better grease/soil tolerance; remains BOV-sprayable. **pH:** 5.8–6.0 **Viscosity:** ≈250–450 cP

Notes: Hydrotrope improves clarity/tolerance; tiny Propylene Carbonate 0.5–1.0% can aid clarity if hazy.

F3 — Heavy-Duty Anionic System

Ingredient	% (w/w)	1 L Batch (g)
Deionized Water	q.s. to 100%	

AOS■40 (Sodium C14■16 Olefin Sulfonate)	8.0%	80 g
CAPB	4.0%	40 g
SCI	2.0%	20 g
Capryl Glucoside	2.0%	20 g
SXS (40%)	2.0%	20 g
HEC (Hydroxyethyl Cellulose, optional)	0.20%	2 g
Sodium Citrate	0.20%	2 g
Chelator (Phytate 0.2% or Gluconate 0.3%)	0.20–0.30%	2–3 g
Polysorbate 20	0.8%	8 g
Peppermint EO	0.25–0.30%	2.5–3 g
Preservative	as F1	10 g (Optiphen) / 8 g (Phenoxy)
Citric Acid (10% solution) to pH	5.8–6.0	

Goal: Stronger cleaning; clearer rinse; anionic■forward. **pH:** 5.8–6.0 **Viscosity:** ≈300–600 cP

Notes: Optional HEC for body; hydrate fully. Keep shear low; filter before BOV.

F4 — Advanced (Builders + Polymer Anti■redemption)

Ingredient	% (w/w)	1 L Batch (g)
Deionized Water	q.s. to 100%	
AOS■40	7.0%	70 g
CAPB	5.0%	50 g
Capryl Glucoside	2.0%	20 g
SCI	2.0%	20 g
SXS (40%)	2.0%	20 g
Propylene Carbonate	1.0%	10 g
DPG	1.5%	15 g
Sodium Polyacrylate	0.20%	2 g
Chelator (Phytate 0.2% or Gluconate 0.3%)	0.20–0.30%	2–3 g
Polysorbate 20	0.8%	8 g
Peppermint EO	0.25–0.30%	2.5–3 g
Preservative	as F1	10 g (Optiphen) / 8 g (Phenoxy)
Citric Acid (10% solution) to pH	5.8–6.0	

Goal: Highest performance; watch clarity/viscosity & filtration. **pH:** 5.8–6.0 **Viscosity:** ≈300–600 cP

Notes: Add polyacrylate <35°C; ensure full hydration; fine filtration (100 µm) pre■BOV.